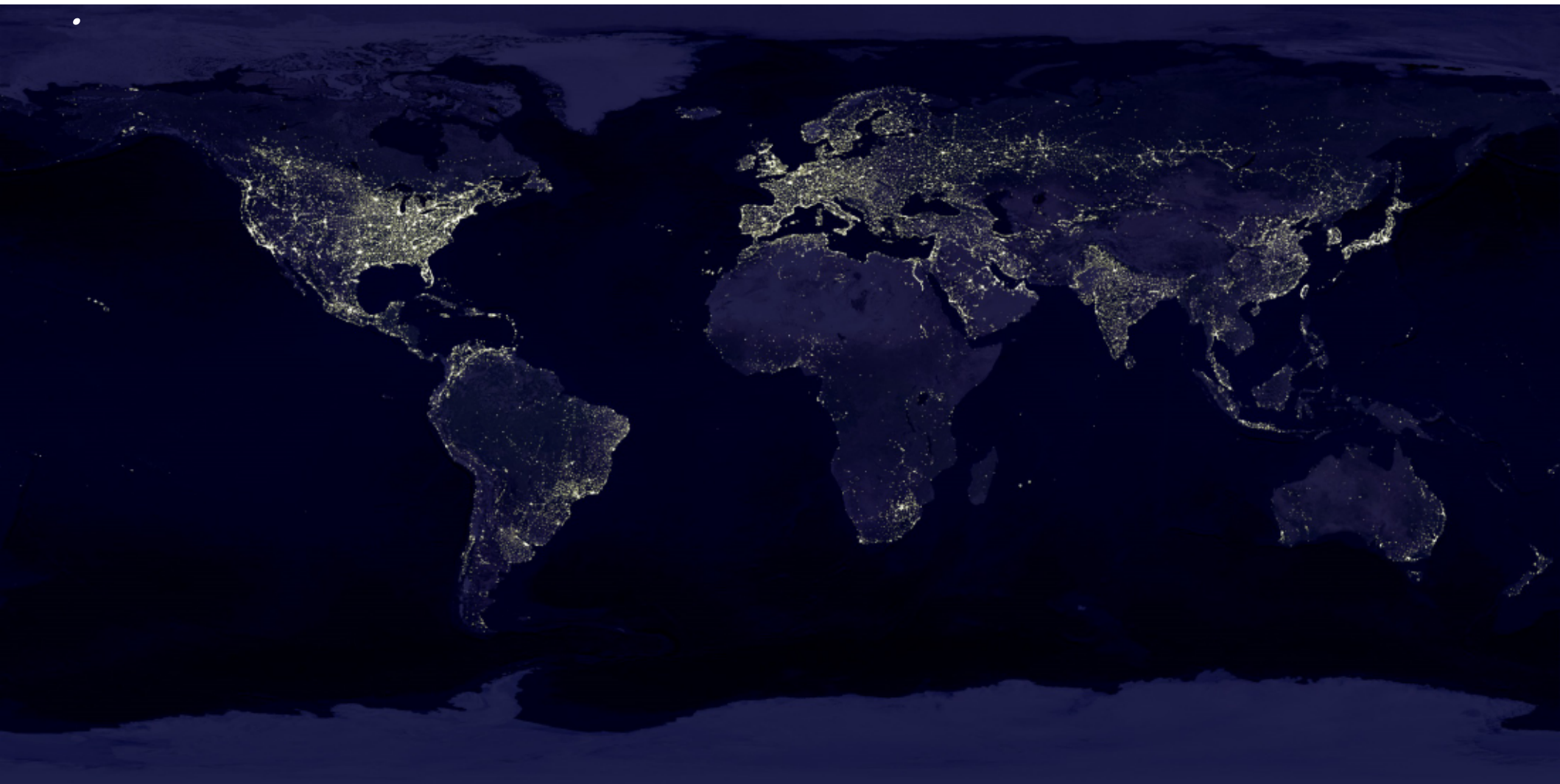


NET- CENTRIC COMPUTING



Dr. Alhassan Adamu



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Course Overview

- Net-Centric Computing typically focuses on the principles, technologies, and methodologies that enable distributed computing and networking systems to function effectively over the internet.
- This field often includes concepts such as cloud computing, distributed systems, communication protocols, and security.



NUC Benchmark (BMAS)

NET-CENTRIC COMPUTING (3 Units)

Distributed Computing,
Mobile & Wireless computing,
Network Security;
Client/Server Computing (using the web),
Building Web Applications.



Course plan

- **Communication and Networking:**

- Basic concepts of networking, circuit switching and packet switching,
- 7-layer ISO model, TCP/IP,
- Connection oriented and connectionless paradigms,
- Physical layer, data link layer, network layer and transport layer services.

- **Network Architectures:**

- Local area networks
- wide area networks,
- Internetworking devices: repeaters, routers, bridges, gateways, wireless networks.



Course plan cont....

Client-server computing and web technologies:

Web Technologies,
Characteristics of web servers and clients,
stateless and stateful client-server interactions,
web protocols,
web-site creation and management,
publishing information and applications

Network Management:

Security issues, security protocols and infrastructure,
Authentication mechanisms, QOS issues, capacity
Planning and performance issues,
Performance evaluation



What is Net-Centric Computing ?

“Distributed environments where applications and data are exchanged among peers across a network on as as-needed basis.”

Applications on distributed environments ?

- more than one machine / one user.

- multiple platform is possible.

How can we exchange data ?

- standard data formats.



What is Net-Centric Computing ?

“Distributed environments where applications and data are exchanged among peers across a network on an as-needed basis.”

Are you my peers ?

security issues.

How soon is as-needed ?

Real-time / batch data exchanging.

Push / Pull.



What is NCC, really ?

Information system (applications).

Multiple machines.

Over the network.

Seamlessly integrated.

Using standard data formats.



Why NCC ?

Stand alone system is not enough

Collaborative among users

All works are workflows by nature.

Pool of resources

Share disks, tapes backup, printers.

Centralized management

Application codes.

Data consistencies.

And much more...



What will be taught in this class ?

Network application architectures and services

A touch on “here-and-there”.

Application-level protocols.

Network-level protocols (TCP/IP)



Fundamentals of Computer Networks



Why Learn about Networking?

Indispensable part of modern society

Commercial – e-commerce, banking, inventorying, telecommunications, archiving, health

Social – critical infrastructure, homeland security, policing

Human interaction/communication – email, chat, videoconferencing, social networking, entertainment

Appears in every facet of engineering

Modern trend – Network every (electronic) device (computers, phones, sensors, planes, cars, TVs, appliances, heart monitors, ...)

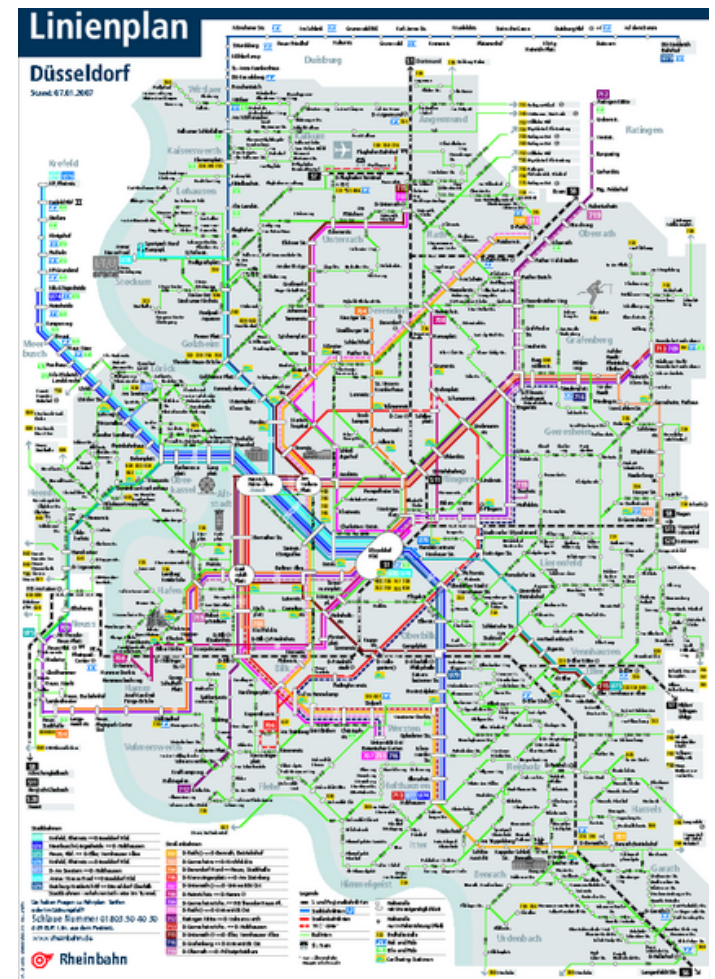


Definition of a Network

A system that carries **a commodity** between 2 or more entities

Examples: Transportation network, electric grid, postal, water, telephone

Computer network: A system that carries information between 2 or more entities, **in the form of electric signals**



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Transportation vs. Computer Networks

Transportation Network

Vehicles/People

Street address

Intersection

Street, highway, path

Link/broadband/path

Traffic jam

Stop and go traffic light

Taking alternative path

Collision

HOV lane

Following a route to school

...

Computer Network

Packets/Payload

IP address

Bridge/router

Network congestion

Flow control

Alternative route

Collision of packets

Flow Priority

Routing algorithm

...



Most commonly known Networks

The Internet*	
Ethernet (LAN)	
WiFi	
3G/4G/5G/6G	
An internet**	

* The global network adopting the IP technology

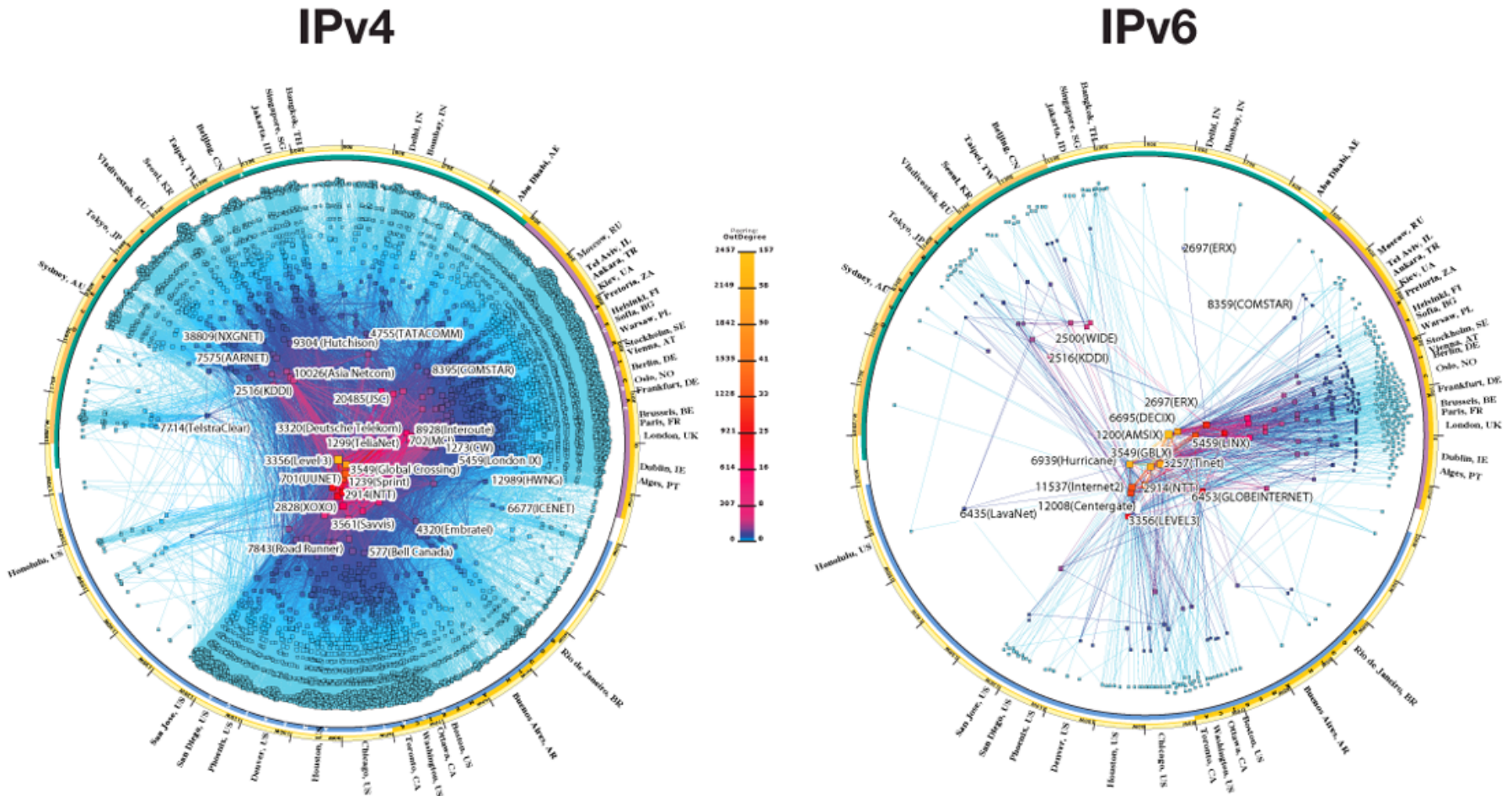
**Internet: A network of networks



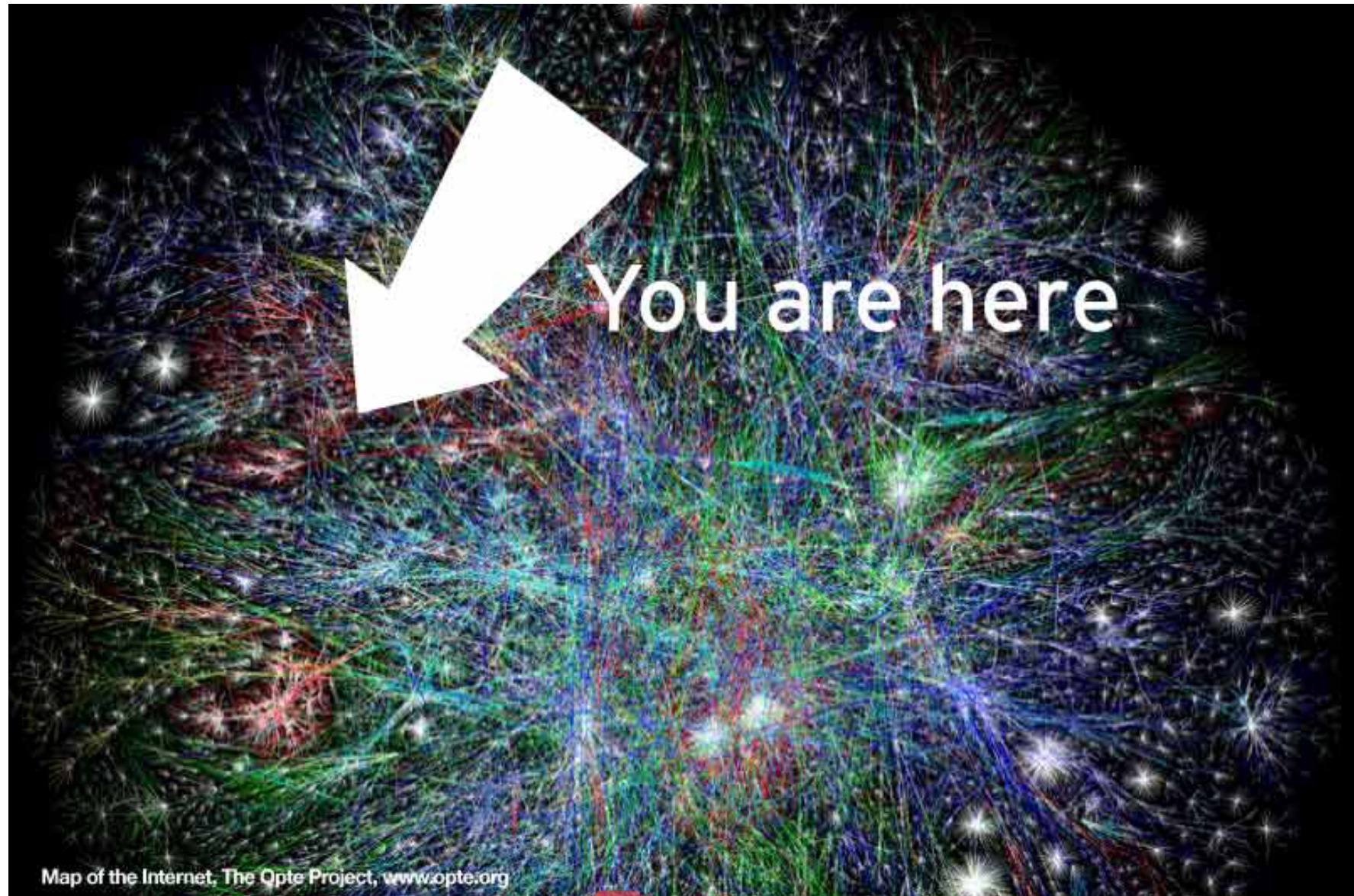
How does the Internet Look Like?

IPv4 & IPv6
INTERNET TOPOLOGY MAP
JANUARY 2009

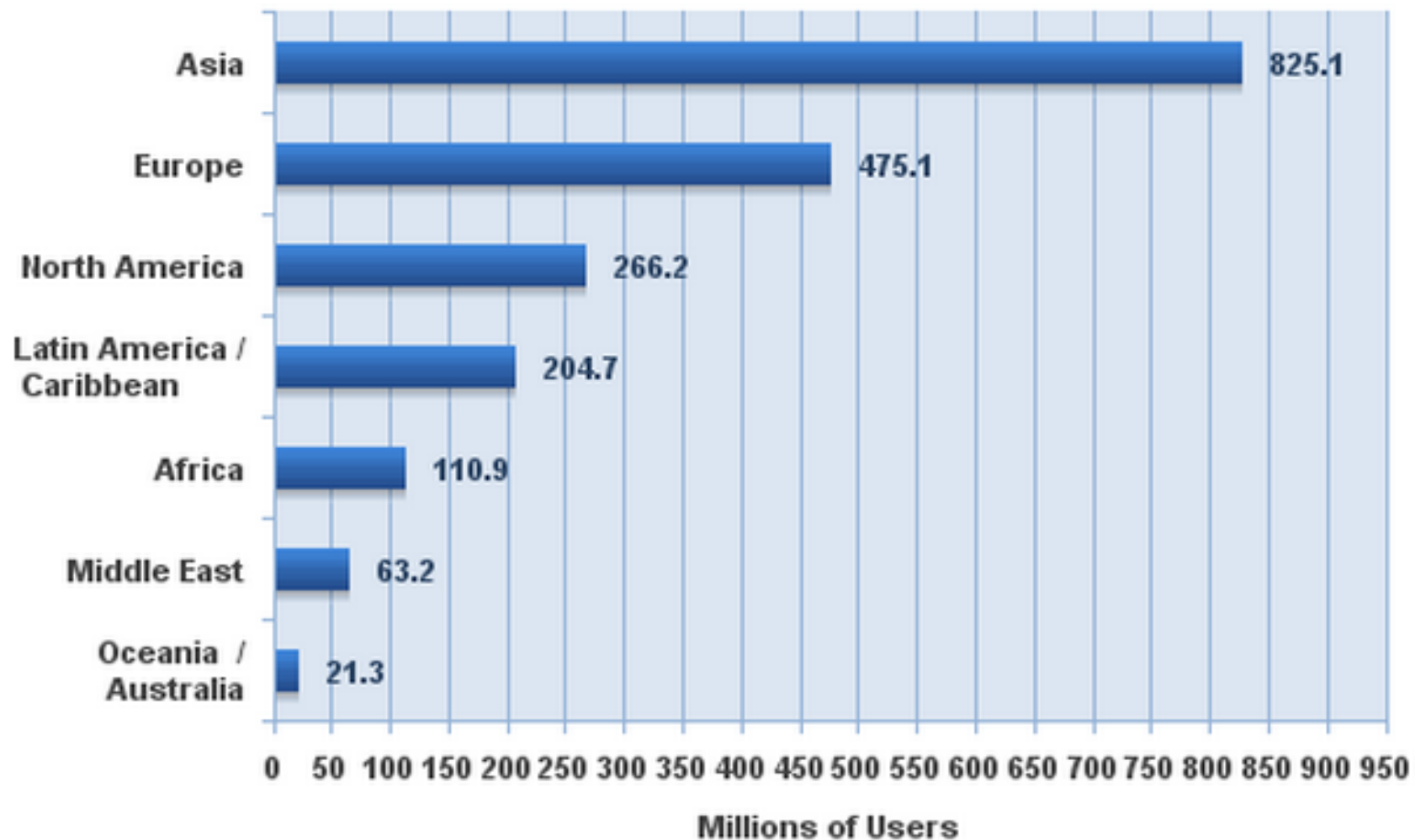
AS-level INTERNET GRAPH



How does the Internet Look Like?



How Many Users?



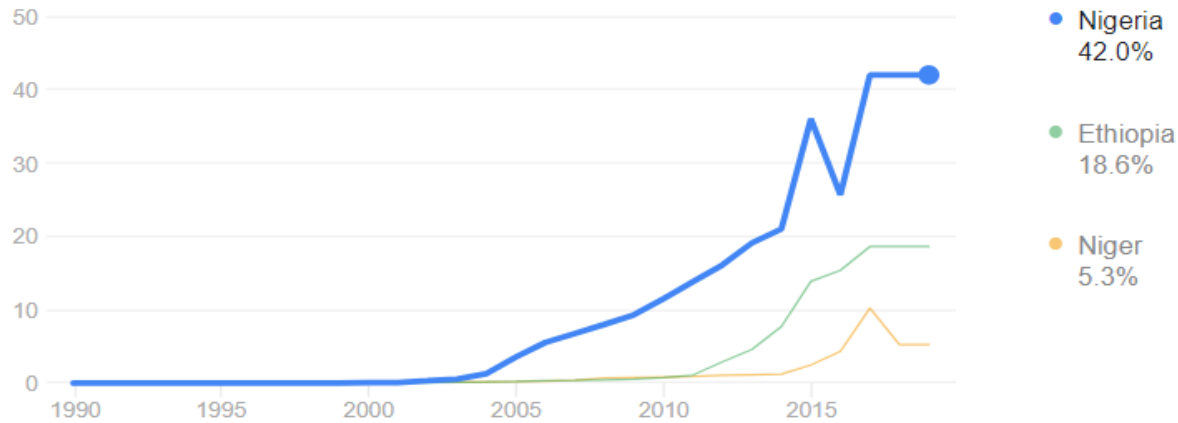
Source: Internet World Stats - www.internetworldstats.com/stats.htm
Estimated Internet users are 1,966,514,816 on June 31, 2010
Copyright © 2010, Miniwatts Marketing Group



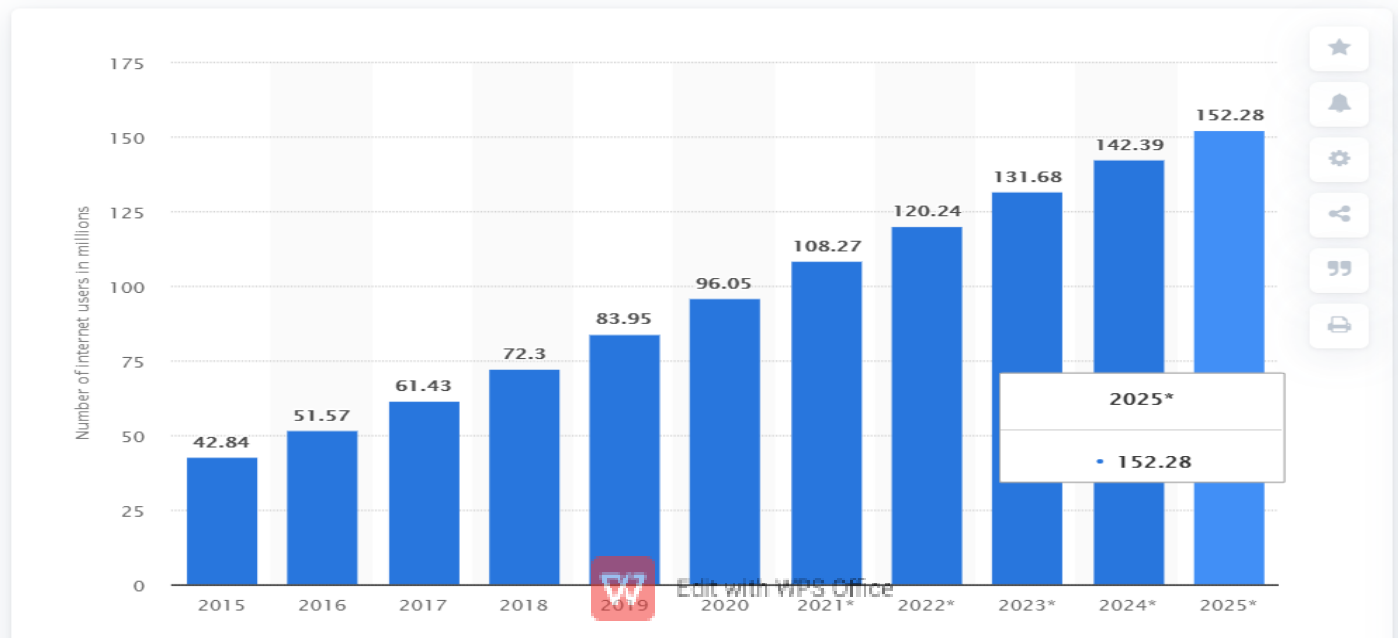
How many more Users?

Nigeria / Internet users

42.0% of the population (2019)



Number of internet users in Nigeria from 2015 to 2025
(in millions)



How much Traffic?

Consumers Leading Way –
58% CAGR in IP Traffic, 2005E-2011E

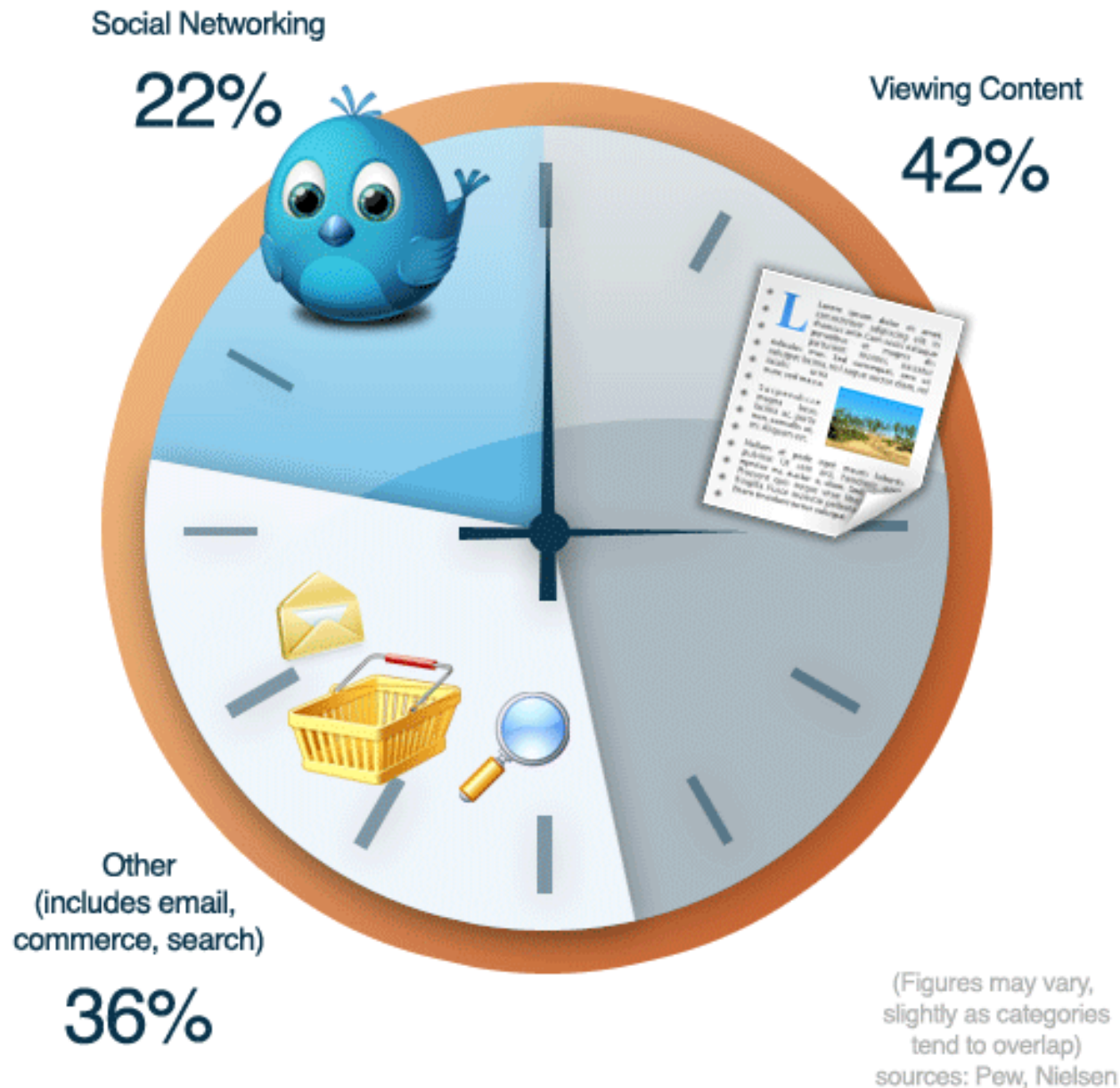
Global IP Traffic



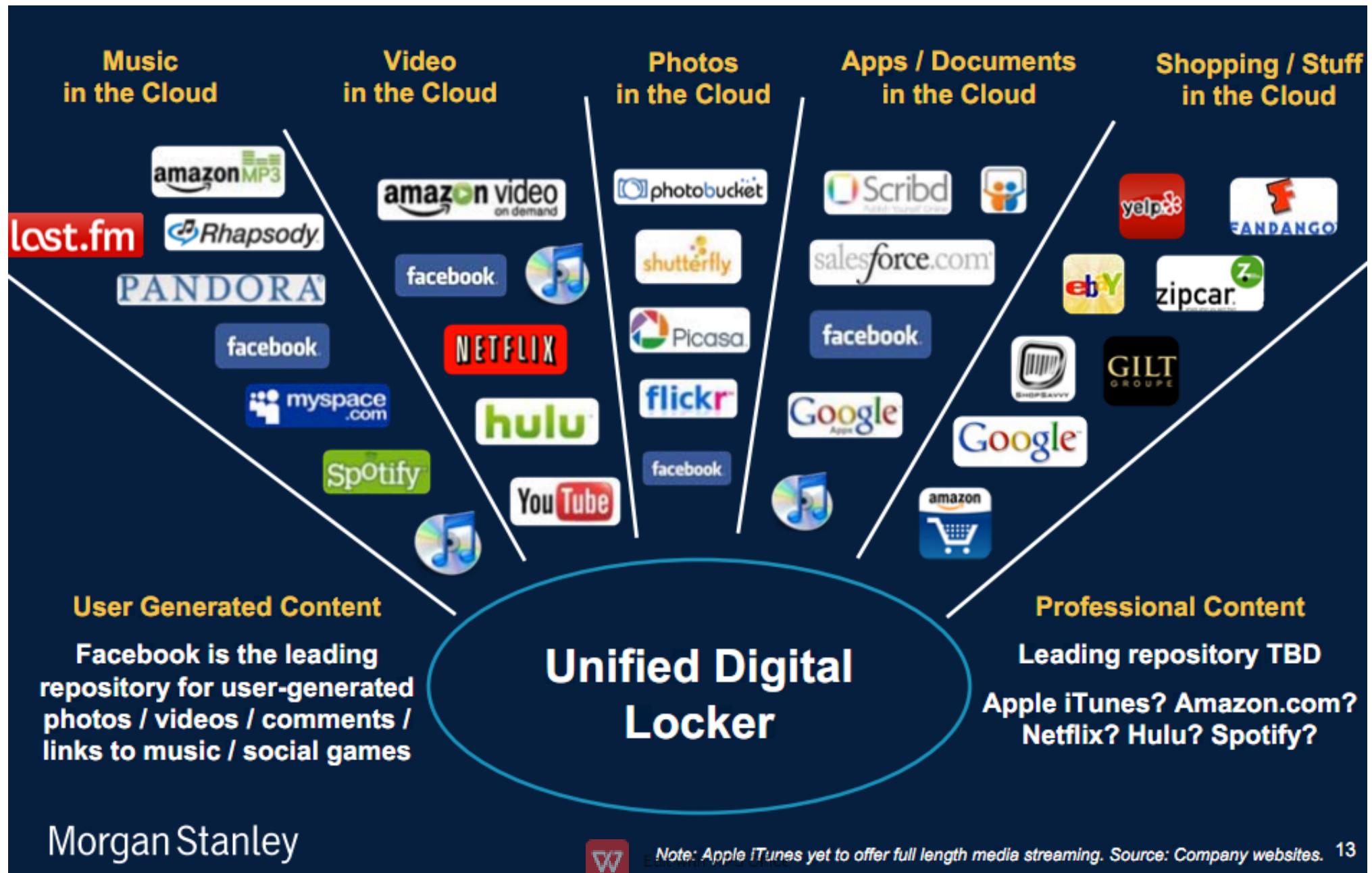
Source: Cisco Systems, Global IP Traffic Forecast and Methodology; Mobility segment (0.1% of traffic in 2007) not displayed



How is Time Spent?



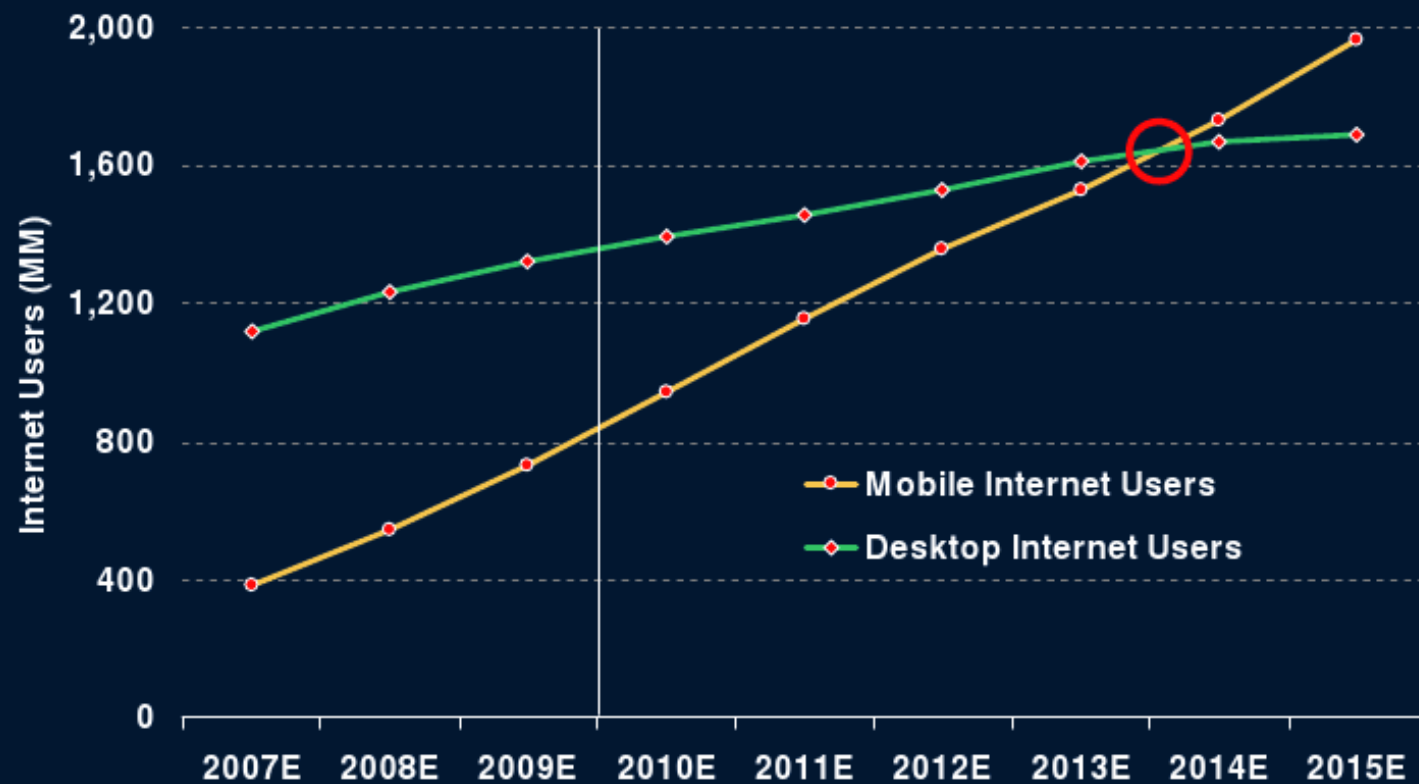
What Do Users Expect?



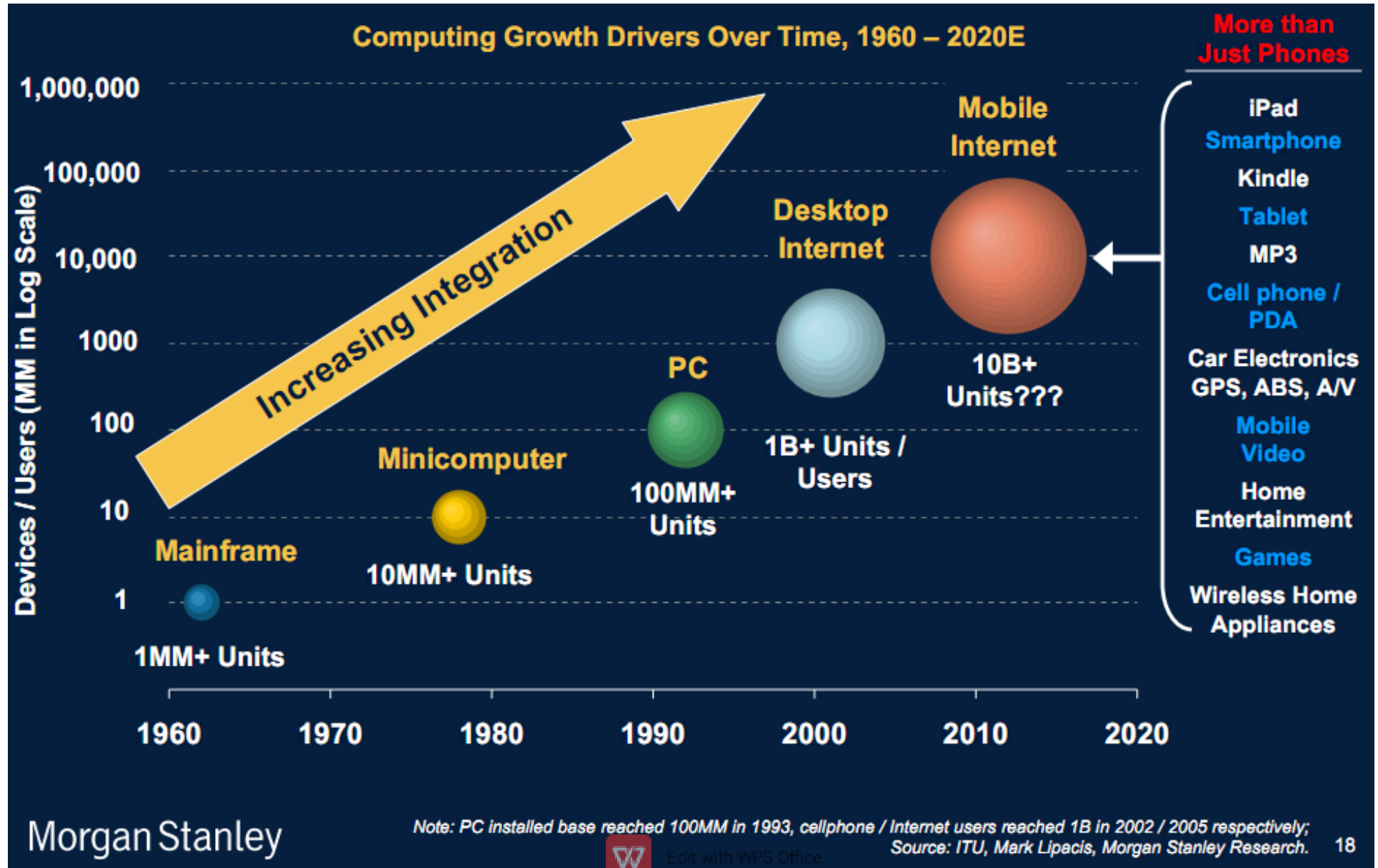
How do they get it?

Mobile Users > Desktop Internet Users
Within 5 Years

Global Mobile vs. Desktop Internet User Projection, 2007 – 2015E



Where are we headed?



Biggest Internet Challenge

Scale

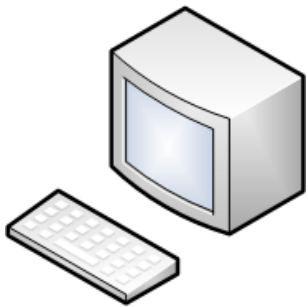
How to manage such a **large system**,
growing rapidly and **uncontrollably**,
consisting of **heterogeneous** devices,
managed by **multiple entities**
having **limited resources**

Let's take things one at a time

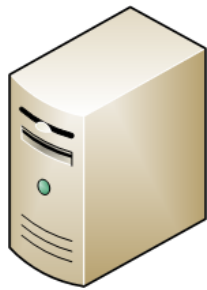


Network Elements

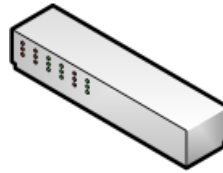
Nodes: Special purpose devices



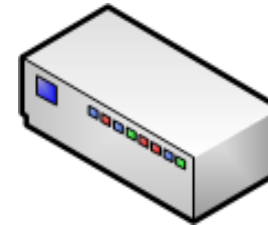
PC



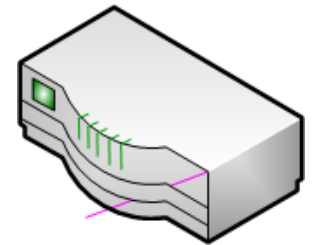
server



switch

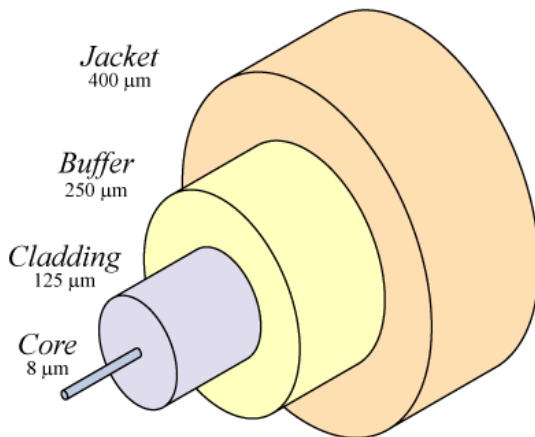


bridge

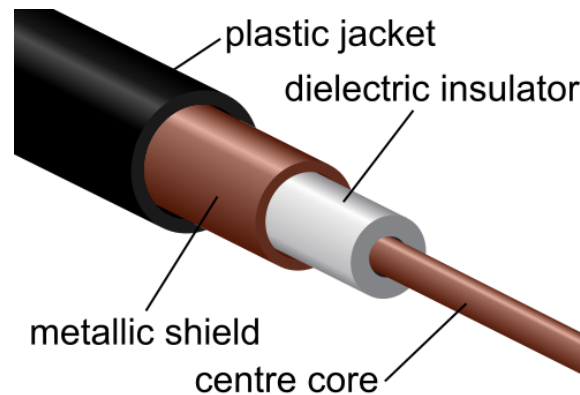


router

Links: Connections between nodes



Optical fiber



Coaxial cable



wireless

Network Design

The task of **connecting nodes via links**, so that nodes can exchange information, reliably, timely, efficiently, safely, privately, “greenly”, and with low cost.

Need to define the network architecture, protocols, applications, interfaces, policies, usages.

Let's start with the architecture

- Directly connected networks

- Circuit-switched networks

- Packet-switched Networks



What Drives Network Design?

Applications

WWW, email, chat, videoconferencing, e-commerce, audio/video streaming, VOIP, file sharing

Who deploys the network

Enterprise, government, end-user

Where is the network deployed

Home, building, campus, state, country, continent, globe



How do we Evaluate a Network

Metrics (think again a transportation network)

How many cars can it service (**throughput**)?

How fast can it service them (**delay**)?

How reliable can it service them (**collisions, losses, outage probabilities, etc**)?

Can it provide any guarantees (QoS)?

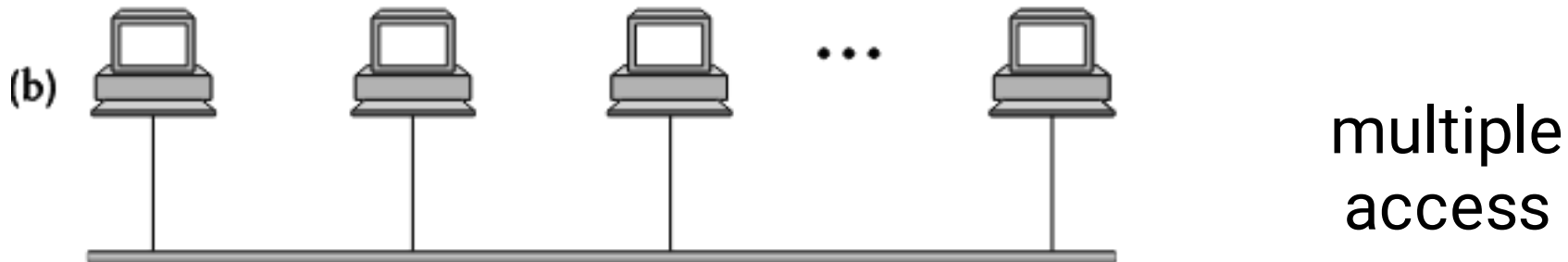
Any other metrics you can think of?



Directly-Connected Networks

Point-to-point links: Each node is directly connected to all others via a link

Multiple access: All nodes share the same physical medium



Switched Networks

Circuit-Switched

A dedicated circuit is established across a set of links

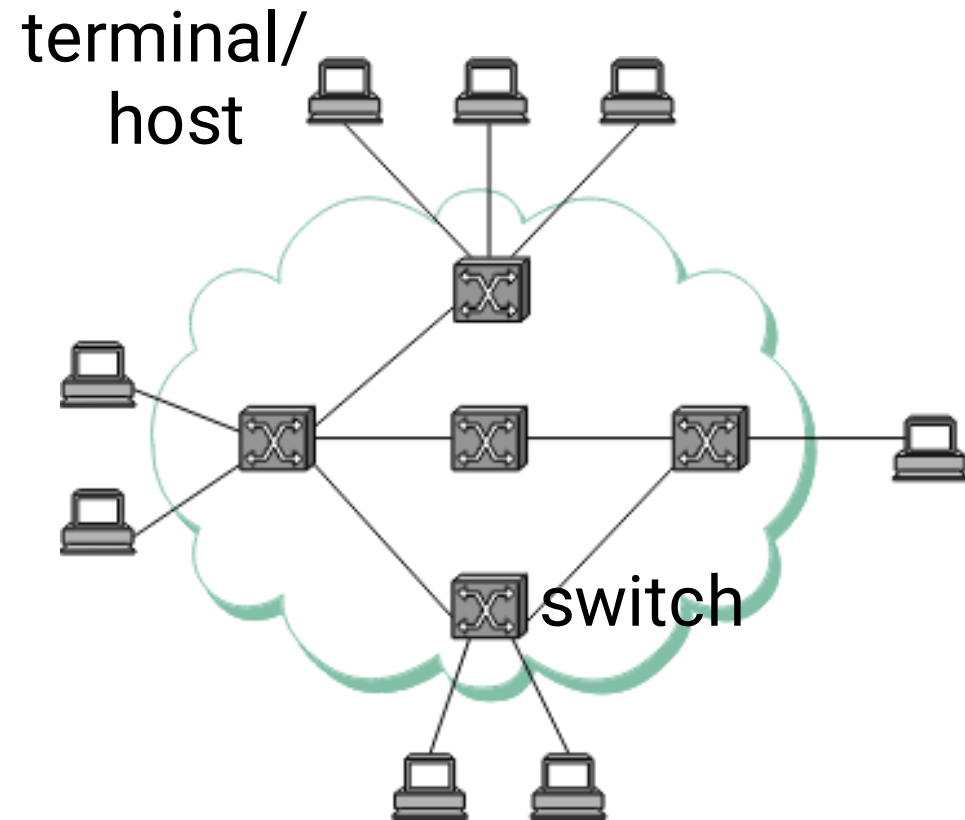
Example: Telephone network

Packet-Switched

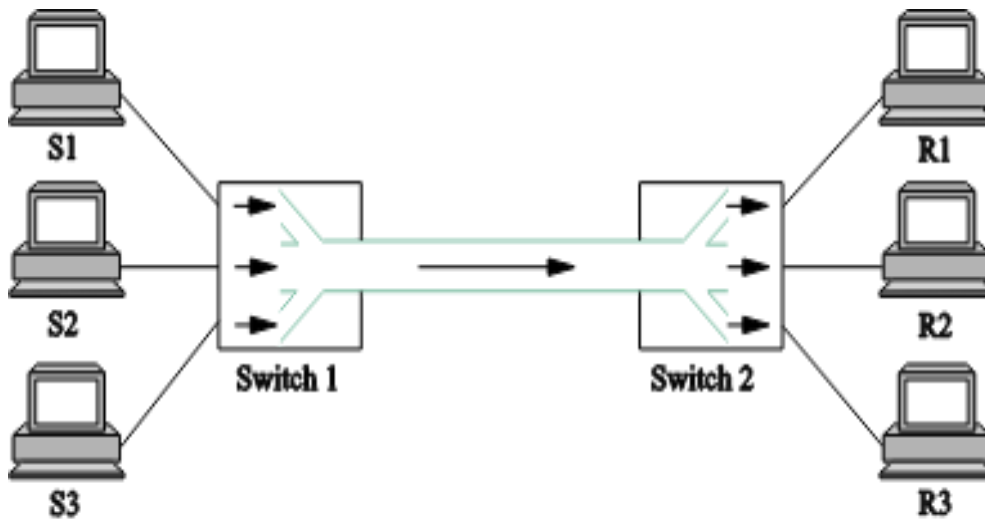
Data is split into blocks called packets or messages.

Store-and-forward strategy

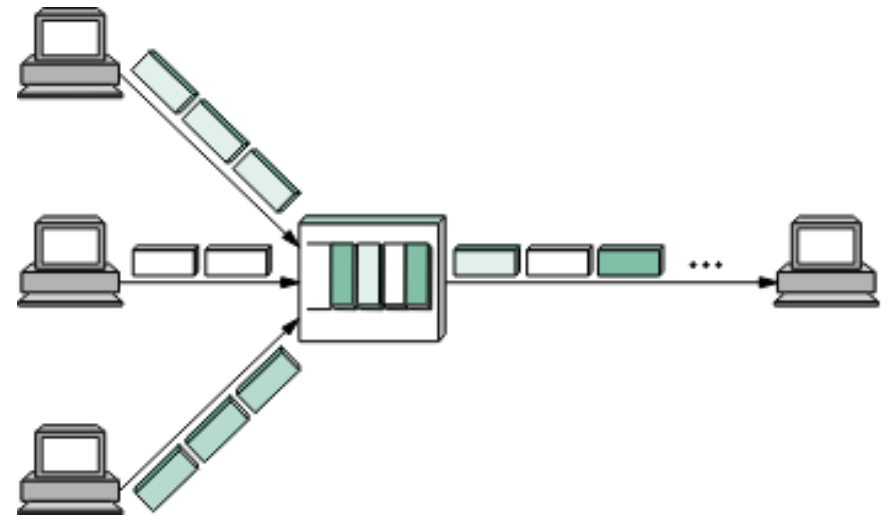
Switches: Store and forward packets



Multiplexing



Three pairs of senders/receivers share the same physical link to communicate



A switch is multiplexing packets from different senders into one packet stream

“Security”

Most of computer science is concerned with *achieving desired behavior*

Security is concerned with preventing undesired behavior

Different way of thinking!

An enemy/opponent/hacker/adversary who is actively and maliciously trying to circumvent any protective measures you put in place



Fortunately, we are winning the security battle

Strong cryptography

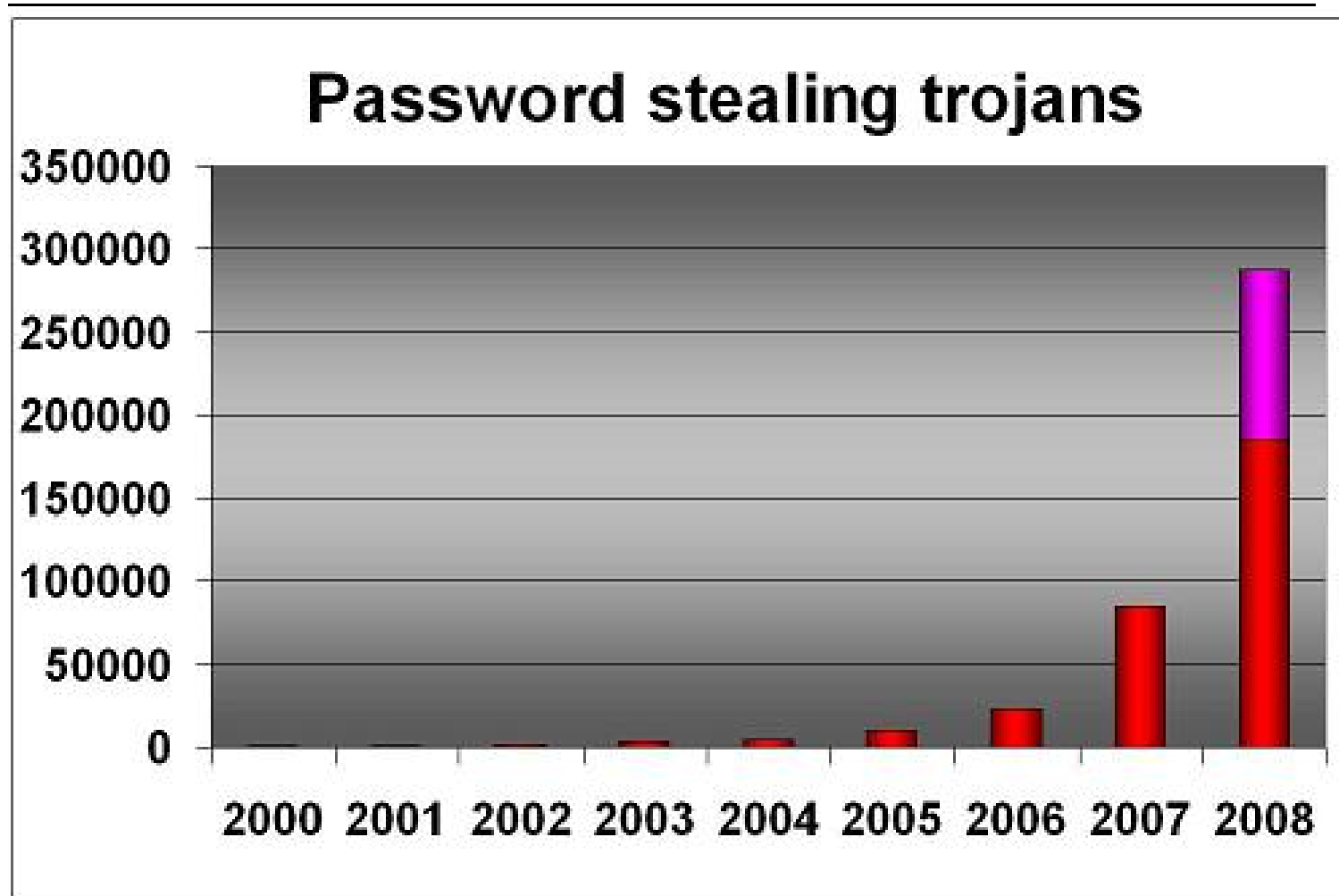
Firewalls, intrusion detection, virus scanners

Buffer overflow detection/prevention

User education



Really??!



A naïve view

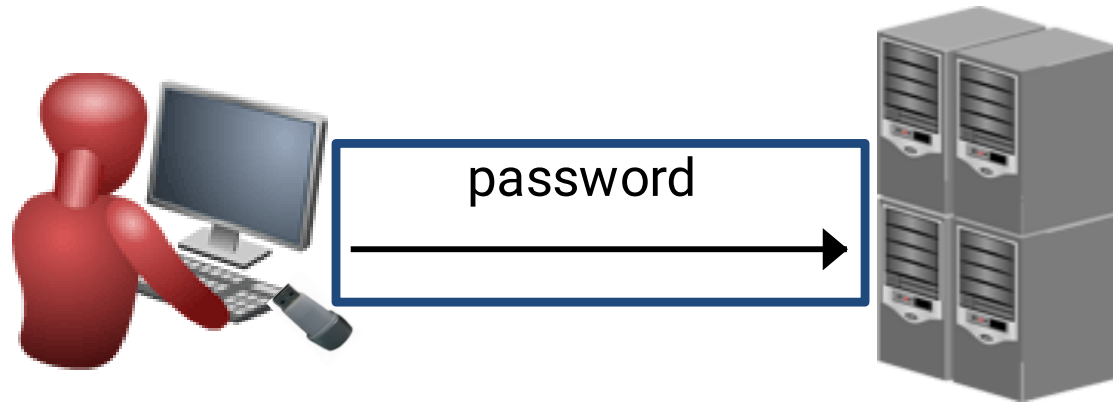
Computer security is about CIA:

Confidentiality, integrity, and availability

These are important, but security is about much more...

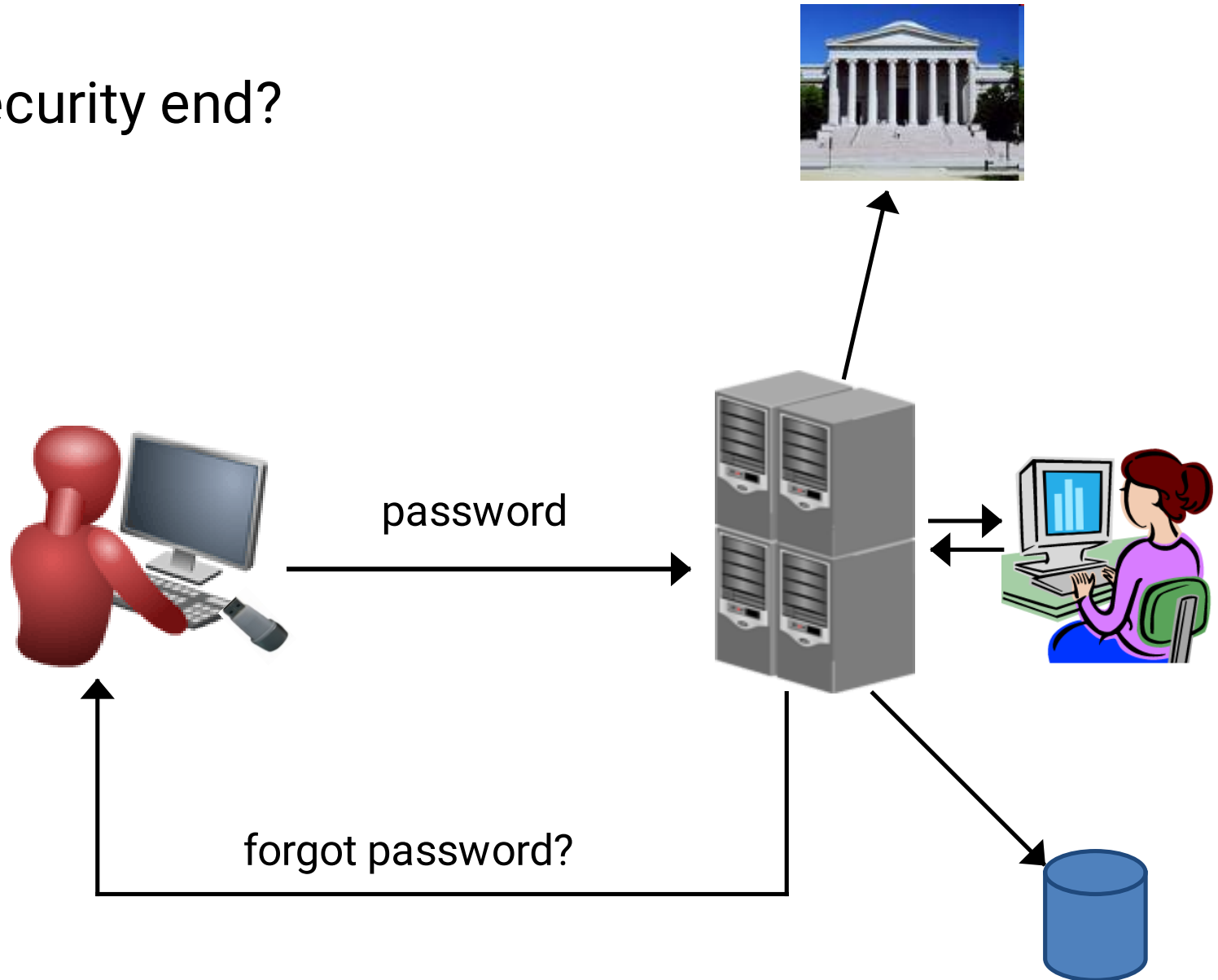


A naïve view



In reality...

Where does security end?



One good attack

Use public records to figure out someone's password

Or, e.g., their SSN, so can answer security question...

The problem is *not* (necessarily) that SSNs are public

The problem *is* that we “overload” SSNs, and use them for more than they were intended

Note: “the system” here is not just the computer, nor is it just the network...



A naïve view

Achieve “absolute” security



In reality...

Absolute security is easy to achieve!

How...?

Absolute security is impossible to achieve!

Why...?

Good security is about *risk management*



Security as a trade-off

The goal is not (usually) “to make the system as secure as possible” ...

...but instead, “to make the system as secure as possible *within certain constraints*” (cost, usability, convenience)

Must understand the existing constraints

E.g., passwords...



Cost-benefit analysis

Important to evaluate what level of security is necessary/appropriate

- Cost of mounting a particular attack vs. value of attack to an adversary

- Cost of damages from an attack vs. cost of defending against the attack

- Likelihood of a particular attack

Sometimes the best security is to make sure you are not the easiest target for an attacker...



“More” security not always better

“No point in putting a higher post in the ground when the enemy can go around it”

Need to identify the *weakest link*

Security of a system is only as good as the security at its weakest point...

Security is not a “magic bullet”

Security is a process, not a product



Computer security is not just about security

Detection, response, audit

How do you know when you are being attacked?

How quickly can you stop the attack?

Can you identify the attacker(s)?

Can you prevent the attack from recurring?

Recovery

Can be much more important than prevention

Economics, insurance, risk management...

Offensive techniques

Security is a process, not a product...



Why is computer security so hard?

Computer networks are “systems of systems”

Your system may be secure, but then the surrounding environment changes

Too many things dependent on a small number of systems

Society is unwilling to trade off features for security

Ease of attacks

Cheap

Distributed, automated

Anonymous

Insider threats

Security not built in from the beginning

Humans in the loop...

Computers ubiquitous...



Computers are everywhere...

...and can always be attacked

Electronic banking, social networks, e-voting

iPods, iPhones, PDAs, RFID transponders

Automobiles

Appliances, TVs

(Implantable) medical devices

Cameras, picture frames(!)

See <http://www.securityfocus.com/news/11499>



“Trusting trust”
(or: how hard *is* security?)



“Trusting trust”

Consider a compiler that embeds a trapdoor into anything it compiles

How to catch?

- Read source code? (What if replaced?)

- Re-compile compiler?

What if the compiler embeds the trojan code whenever it compiles a compiler?

- (That's nasty...)



“Trusting trust”

Whom do you trust?

Does one really need to be this paranoid??

Probably not

Sometimes, yes

Shows that security is complex...and essentially impossible

Comes back to risk/benefit trade-off



END

